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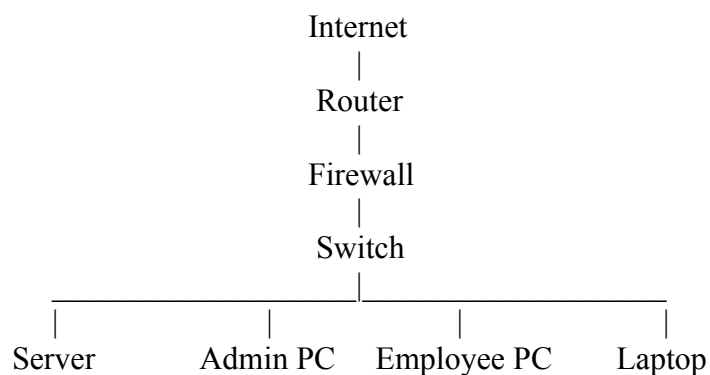
Synchrony Skills Academy

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Network Security 1

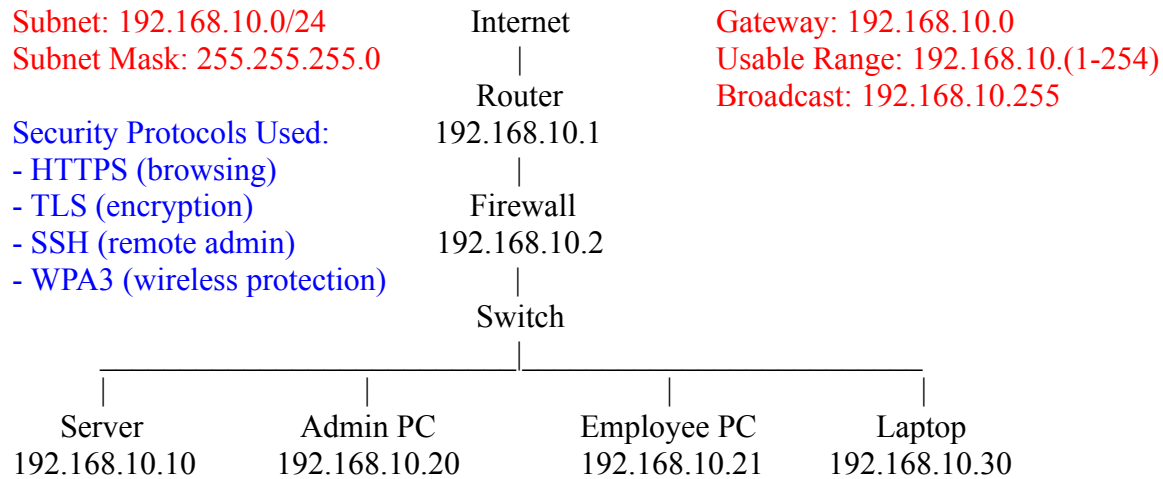
Understand and Implement Network Topologies:

LAN Network Topology Diagram



This LAN topology supports secure communication and network management because all traffic flows through the router, firewall, and network switch before reaching any computers on the LAN. The firewall filters traffic and enforces security rules, while the switch and WAP control how the devices communicate with each other and communicate with the internet. This centralized structure also makes the network easier to manage because administrators can monitor traffic and manage connected devices at the router level or just on a per device basis.

Design Network Protocols and Architectures:



Employee PC:	
OSI Model:	TCP/IP Model:
Layer 7 - Application Using a browser to request an HTTPS page. Layer 6 - Presentation Browser formats data for display & uses TLS. Layer 5 - Session PC establishes a communication session with the remote web server. Layer 4 - Transport TCP breaks the webpage data into segments and delivers it to the destination. Layer 3 - Network IP assigns source and destination addresses so the request can travel to the web server. Layer 2 - Data Link Ethernet or Wi-Fi moves data across the LAN and uses MAC addresses to deliver it to the next device Layer 1 - Physical Data travels as electrical signals/radio waves.	Application Layer The employee's browser uses HTTPS to request and receive the web content from the website. Transport Layer TCP creates a connection between the PC and the server so that the webpage data arrives in the correct order. Internet Layer IP handles addressing and routing so that the request can move from the local network to the remote website across the internet. Network Access Layer Ethernet or Wi-Fi carries the data across the local network to the router or the access point before it leaves to the internet.

Implement Network Security Fundamentals:

Step 1: Install Suricata & iptables

```
sudo apt update

sudo apt install suricata suricata-update

iptables-persistent -y
```

Step 2: Implement 1 Firewall Rule

```
sudo iptables -A INPUT -p tcp --dport 23 -j DROP

sudo iptables -L -n -v

sudo sh -c "iptables-save > /etc/iptables/rules.v4"
```

(This rule tells Kali Linux to drop inbound TCP traffic on port 23.)

Step 3: Implement 1 IDS Configuration

3a. Identify the Interface Using `ip a`.

Example: Interface is `eth0` and the local subnet is `192.168.56.0/24`.

3b. Edit the Suricata Configuration File by Typing:

```
sudo nano /etc/suricata/suricata.yaml
↓
vars:
  address-groups:
    HOME_NET: "[192.168.56.0/24]"

af-packet:
  - interface: eth0
```

3c. Create a Local Rule by Typing:

```
sudo nano /etc/suricata/rules/local.rules
```

↓

```
alert icmp any any -> $HOME_NET any (msg:"LAB ICMP  
ping detected"; sid:1000001; rev:1;)
```

3d. With nano, Make Sure the File is Loaded in suricata.yaml:

```
sudo nano /etc/suricata/suricata.yaml
```

↓

```
rule-files:  
    - suricata.rules  
    - local.rules
```

3e. Update the Ruleset and Test the Configuration:

```
sudo suricata-update
```

```
sudo suricata -T -c /etc/suricata/suricata.yaml -i  
eth0
```

3f. Start Suricata in IDS Mode:

```
sudo suricata -c /etc/suricata/suricata.yaml -i eth0
```

Step 4: Implement 1 IPS Configuration**4a. Edit the Rules File Again:**

```
sudo nano /etc/suricata/rules/local.rules
```

↓

```
drop icmp any any -> $HOME_NET any (msg:"LAB IPS  
blocked ICMP ping"; sid:1000002; rev:1;)
```

4b. Update the Ruleset and Test the Configuration Again:

```
sudo suricata-update
```

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

4c. Send ICMP Traffic to NFQUEUE with iptables:

```
sudo iptables -I INPUT -p icmp -j NFQUEUE --queue-num  
0
```

4d. Start Suricata in inline IPS mode:

```
sudo suricata -q 0 -c /etc/suricata/suricata.yaml
```

Step 5: How to Get 1 Example of Detected Events**5a. From another Kali Linux VM on the same network, send a ping:**

```
ping [ip-of-the-kali-linux-machine]
```

5b. Collect the example by typing:

```
sudo tail -f /var/log/suricata/fast.log
```

Implement Access Control Measures:

Step 1: Create 2 Example Users on Kali Linux and Configure the Access Level

```
sudo adduser employee1  
  
sudo adduser admin1  
  
sudo usermod -aG sudo admin1
```

Step 2: Create a Folder and File that will be Used to Demonstrate Access Control

```
sudo mkdir /securefiles  
  
sudo touch /securefiles/confidential.txt  
  
sudo chown admin1:admin1 /securefiles/confidential.txt  
  
sudo chmod 640 /securefiles/confidential.txt
```

Access Control Model Used: Discretionary Access Control (DAC)

↑ (The file owner can decide which users or groups are allowed to read, write, or modify the file.)

Step 3: Document the Normal Linux Permissions First

```
ls -l /securefiles/confidential.txt  
  
↓  
  
-rw-r----- 1 admin1 admin1 0 Sep 20 10:00  
  
/securefiles/confidential.txt
```

Step 4: Configure 1 ACL

4a. Give employee1 Read-only Access to the File, Even Though They are Not the

Owner:

```
sudo setfacl -m u:employee1:r--  
/securefiles/confidential.txt
```

4b. Verify the ACL:

```
getfacl /securefiles/confidential.txt
```

Secure Wireless Networks:**Step 1: Install Required Tools**

1a. Install kismet and hostapd by typing:

```
sudo apt update  
sudo apt install hostapd kismet -y
```

1b. Verify installation by typing:

```
hostapd -v  
kismet --version
```

Step 2: Configure hostapd

2a. Use nano to create the WPA3-SAE hostapd configuration file:

```
sudo nano /etc/hostapd/hostapd.conf
```

↓

```
interface=wlan0
```

```
driver=nl80211
```

```
ssid=SecureOfficeWiFi
```

```
hw_mode=g
```

```
channel=6
```

```
wpa=2
```

```
wpa_key_mgmt=SAE
```

```
rsn_pairwise=CCMP
```

```
ieee80211w=2
```

```
sae_password=StrongLabPassword123!
```

2b. Edit the hostapd defaults file and set the configuration path:

```
sudo nano /etc/default/hostapd
```

↓

```
DAEMON_CONF="/etc/hostapd/hostapd.conf"
```

3b. Enable and start hostapd:

```
sudo systemctl unmask hostapd  
sudo systemctl enable hostapd  
sudo systemctl start hostapd  
sudo systemctl status hostapd
```

3c. Add a preventive wireless access control measure with an allowlist file. Add approved MAC addresses (one per line). For example:

```
sudo nano /etc/hostapd/hostapd.accept
```

↓

```
00:11:22:33:44:55
```

```
AA:BB:CC:DD:EE:FF
```

3d. Then, edit hostapd.conf:

```
sudo nano /etc/hostapd/hostapd.conf
```

↓

```
macaddr_acl=1
```

```
accept_mac_file=/etc/hostapd/hostapd.accept
```

Step 3: Configure kismet**3a. Start kismet by typing:**

```
sudo kismet
```

3b. Configure Kismet alerts for unauthorized wireless activity. For example:

```
sudo nano /etc/kismet/kismet_site.conf
```

↓

```
alert=APSPOOF,5/min,1/sec
```

Step 4: Check kismet datasource and alert status for evidence of the WIPS functioning:

4a. Kismet datasource status will confirm whether or not the wireless adapter was active, and Kismet alert status will confirm that the alerts are being generated.

Utilize Network Security Tools:**Step 1: Install Required Tools****1a. Update Kali Linux, then install Wireshark, Nmap, and Nikto:**

```
sudo apt update
```

```
sudo apt install wireshark nmap nikto -y
```

2a. Verify the installations:

```
wireshark --version
```

```
nmap --version
```

```
nikto -Version
```

Step 2: Document One Wireshark Capture**2a. Start Wireshark:**

```
sudo wireshark
```

2b. In Wireshark, do the following:

- Select the active network interface
- Start a capture
- Browse a website or ping another host to create traffic.
- Stop the capture after a short period of time.

2c. After capturing the traffic, use various packet filters to analyze the traffic:

- dns
- icmp
- http
- etc.

2d. Save the packet capture file from Wireshark.

Step 3: Document One **Network Vulnerability Scanner Report** with Nmap

3a. Run the following command:

```
nmap -sV --script vuln <targetIp> -oA nmap_vuln_report
```

- `-sV` identifies service versions
- `--script vuln` runs NSE scripts in the vulnerability category
- `-oA` saves the scan in normal, XML, and grepable formats

3b. View/Upload `nmap_vuln_report.nmap` as evidence.

Step 4: Document One **Network Penetration Testing Tool Output** with Nikto

4a. Run the following command:

```
nikto -h http://<target-ip> -Format htm -o report.html
```

- `-h` sets the target host or URL (should be a web server for this tool)
- `-Format htm` saves the report in HTML format
- `-o` writes the scan output to a file

4b. View/Upload `report.html` as evidence.

Monitor and Respond to Network Security Events:

Step 1: Add a Detection Rule

1a. Use the same Suricata configuration from earlier.

1b. Configure a new Suricata rule. Example for a basic ICMP event:

```
sudo nano /etc/suricata/rules/local.rules
```

↓

```
alert icmp any any -> $HOME_NET any (msg:"LAB ICMP  
ping detected"; sid:1000001; rev:1;)
```

Step 2: Test and Start Monitoring

2a. Test the configuration and then start Suricata:

```
sudo suricata -T -c /etc/suricata/suricata.yaml -i  
eth0
```

```
sudo suricata -c /etc/suricata/suricata.yaml -i eth0
```

Step 3: Generate a Network Security Incident

3a. On another host, ping the Kali Linux machine:

Option A:

```
ping <kali-machine-ip>
```

Option B:

```
nmap -sn <kali-ip>
```

Step 4: Review the Logs

4a. You can view either simple or more structured logs:

Simple:

```
sudo tail -f /var/log/suricata/fast.log
```

(The tail command with no options automatically prints out the last 10 lines of the file and the -f in the command will constantly update those 10 lines with current data)

Structured:

```
cat /var/log/suricata/eve.json
```

Step 5: Identify the Incident

5a. For example:

Incident Type: Suspicious network probing/unauthorized network traffic

Severity: Medium to Low

Potential Impact: Reconnaissance and unauthorized scanning

Reason for Classification: The event matched a security detection rule and showed suspicious behavior

Step 6: Create an Incident Response Plan**6a. Example steps:****1. Validate the Alert**

Review the logs to confirm that the traffic was not a false positive.

2. Identify the Source

Check the source IP, destination IP, protocol, and time of the event.

3. Assess the Severity

Determine whether the activity is harmless, accidental, or malicious.

4. Contain the Activity

Block the source IP with a firewall rule or isolate the affected host.

5. Investigate Further

Review additional logs/activity to see if the event is part of a larger attack.

6. Document the Incident

Record the alert details, classification, and the response actions taken.

7. Review all of the Lessons Learned

Determine whether or not the monitoring rules, the firewall policies, or the alert thresholds should be improved.